Native F4/F5, F6 and root solvers

Run 20261019-165506-boolean-recursive-native-188e81, source 0acfc55b7f. Dedicated c7i.2xlarge, one worker and a reserved complete physical core; four-worker controls reserve two cores/SMT siblings. Exact source/compiler/CPU/ISA and binary manifests remain in the raw tree. Baseline: frozen a78 algorithm with all three previously qualified GF2 switches. Candidate: identity-bound prepared construction and bounded quadratic-tail policies.

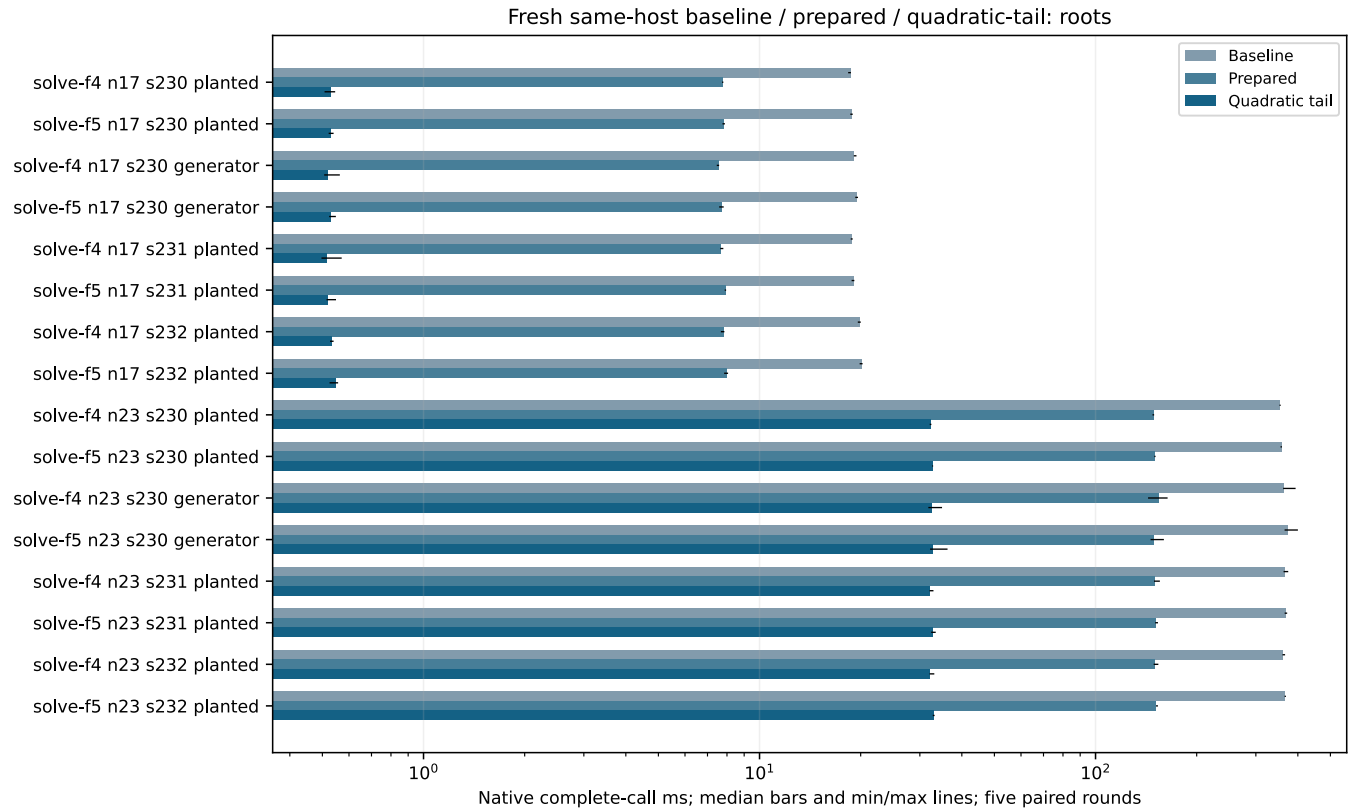
Original n17/n23 fixtures have16/22 Boolean variables on icv1-f2m17-tm101-00378d4e / icv1-f2m23-tm5197-1f85e9e1. Planted seed230 and fixed generator control plus planted holdouts231/232; random=true means generator here. n19's36-variable admission gap remains separate. No fixture or baseline is replaced by the full-DLP standard subspaces.

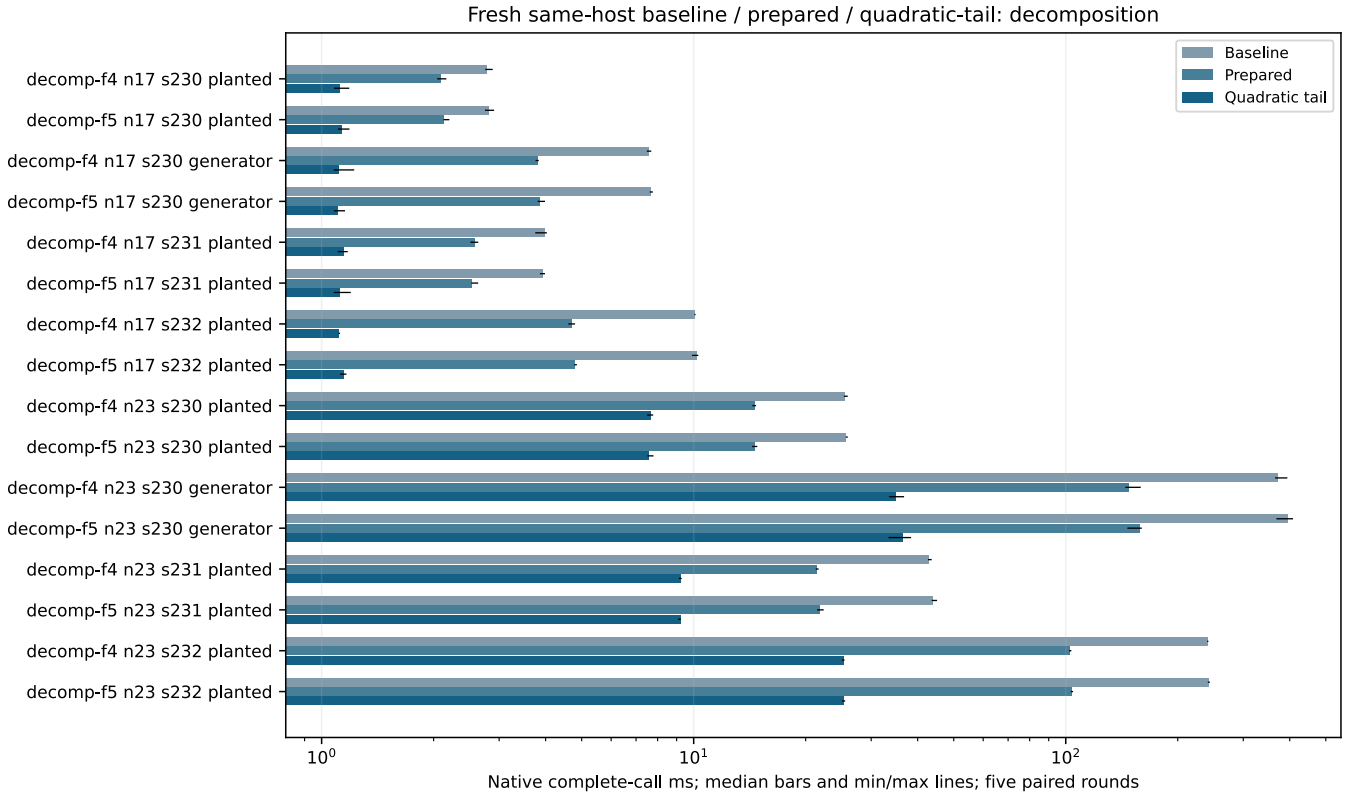
All eighteen arms are rerun with this source's declared policies. Complete original roots and first-witness decomposition remain separate; common curve/base construction is outside component timings. Fresh target ANF, solving, lifting and internal verification are charged to decomposition; Semaev fresh table setup is charged once. Independent original-root/group audits run after elapsed capture.

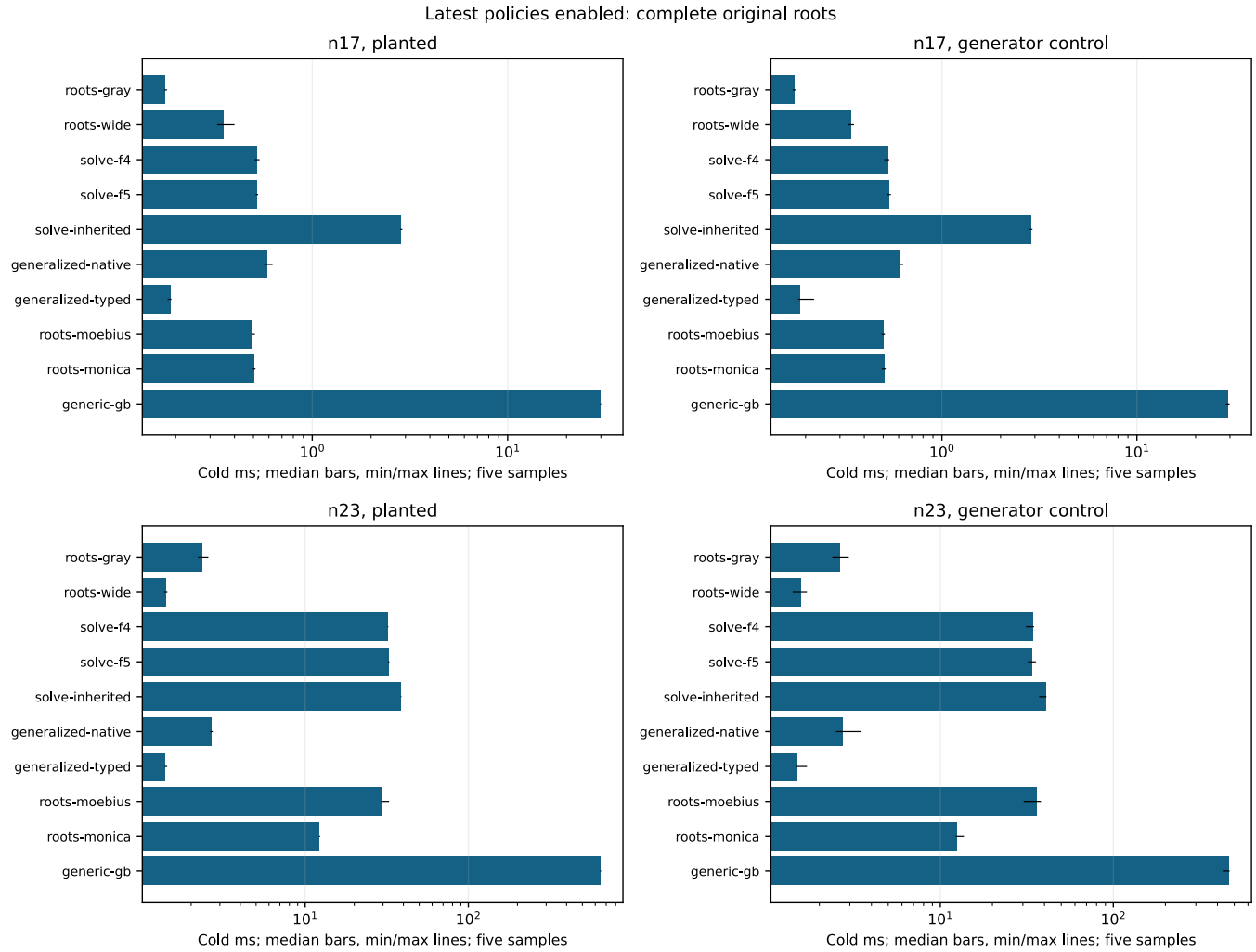
Five paired rounds, first eligible of at most three attempts, exact original outputs, exact3125-resample CI and max(2%, twice A/A noise) gate. Rejected attempts retained: 42. All four multiworker controls neutral: True. Every slower/neutral/missing cell remains in native-summary.json.

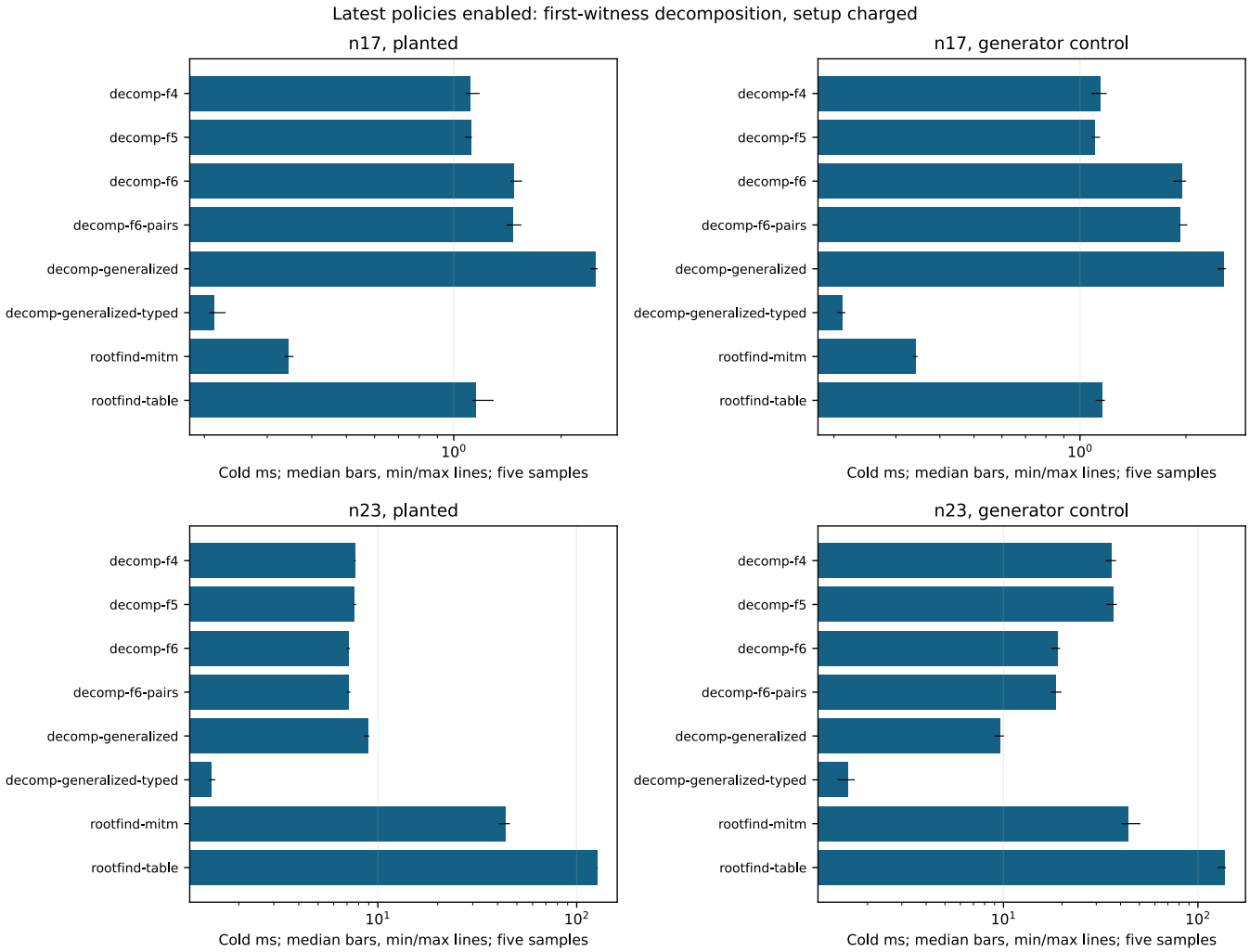
Prepared is a combined active-layout/top-degree/linear-tail/direct-packing ablation. Quadratic-tail retains an algebraic root reduction, then exhaustively completes at most16 free quadratic variables using the existing Gray kernel; original coordinates, root order, caps and verification remain enforced. This is a hybrid engineering change, not a novel enumeration algorithm. Other split rules/custom node oracles keep recursion.

The matrix switches stay opt-in. Source/protocols: 0acfc55b7f, PROTOCOL.md and the frozen runner/protocols in the raw tree. Sources0acfc55b7f,71c4c99d85 and9bed104b0a are separate experiments. The final adaptive variants preserve cheap degree-two refutations below16 original variables; legacy policy names remain for replay. Full-table setup uses the same exact sorted tuples serially under an explicit one-worker resource contract. No cross-host pooling, external-fastest, Arm64/GPU, whole-IC/rho or hard-family8x conclusion follows from component calls.




Fresh same-host baseline / prepared / quadratic-tail: decomposition
Baseline
Prepared
Quadratic tail
decomp-f4 n17 s230 planted
decomp-f5 n17 s230 planted
decomp-f4 n17 s230 generator
decomp-f5 n17 s230 generator
decomp-f4 n17 s231 planted
decomp-f5 n17 s231 planted
decomp-f4 n17 s232 planted
decomp-f5 n17 s232 planted
decomp-f4 n23 s230 planted
decomp-f5 n23 s230 planted
decomp-f4 n23 s230 generator
decomp-f5 n23 s230 generator
decomp-f4 n23 s231 planted
decomp-f5 n23 s231 planted
decomp-f4 n23 s232 planted
decomp-f5 n23 s232 planted
10^0
10^1
10^2
Native complete-call ms; median bars and min/max lines; five paired rounds


Latest policies enabled: complete original roots
n17, planted
n17, generator control
n23, planted
n23, generator control
roots-gray
roots-wide
solve-f4
solve-f5
solve-inherited
generalized-native
generalized-typed
roots-moebius
roots-monica
generic-gb
Cold ms; median bars, min/max lines; five samples


Latest policies enabled: first-witness decomposition, setup charged
n17, planted
n17, generator control
n23, planted
n23, generator control
decomp-f4
decomp-f5
decomp-f6
decomp-f6-pairs
decomp-generalized
decomp-generalized-typed
rootfind-mitm
rootfind-table
10^0
10^1
10^2
Cold ms; median bars, min/max lines; five samples

All paired recursive-solver outcomes

Arm	n	Seed	Target	Before ms	Candidate ms	Ratio	Decision
solve-f4	17	230	P	18.6683	0.5305	35.2518	faster
solve-f5	17	230	P	18.7984	0.5316	35.2952	faster
decomp-f4	17	230	P	2.7783	1.1205	2.4795	faster
decomp-f5	17	230	P	2.8139	1.1352	2.4746	faster
solve-f4	17	230	G	19.1387	0.5177	37.1055	faster
solve-f5	17	230	G	19.4664	0.5320	36.5932	faster
decomp-f4	17	230	G	7.5975	1.1128	6.8504	faster
decomp-f5	17	230	G	7.6835	1.1053	6.9513	faster
solve-f4	17	231	P	18.8223	0.5164	36.6637	faster
solve-f5	17	231	P	19.0790	0.5209	36.5280	faster
decomp-f4	17	231	P	3.9782	1.1500	3.4860	faster
decomp-f5	17	231	P	3.9246	1.1189	3.4978	faster
solve-f4	17	232	P	19.9130	0.5343	37.2596	faster
solve-f5	17	232	P	20.1732	0.5480	36.7255	faster
decomp-f4	17	232	P	10.0959	1.1127	9.0538	faster
decomp-f5	17	232	P	10.1780	1.1489	8.8757	faster
solve-f4	23	230	P	353.2220	32.2922	10.9370	faster
solve-f5	23	230	P	357.6871	32.7721	10.9144	faster
decomp-f4	23	230	P	25.4665	7.6754	3.3198	faster
decomp-f5	23	230	P	25.7097	7.5960	3.3838	faster
solve-f4	23	230	G	363.7493	32.5795	11.3016	faster
solve-f5	23	230	G	373.4116	32.7732	11.3496	faster
decomp-f4	23	230	G	371.3619	34.8908	10.7285	faster
decomp-f5	23	230	G	396.3342	36.5644	10.9111	faster
solve-f4	23	231	P	367.1030	32.2413	11.3808	faster
solve-f5	23	231	P	368.6304	32.8270	11.2185	faster
decomp-f4	23	231	P	42.8434	9.2654	4.6716	faster
decomp-f5	23	231	P	43.9673	9.2237	4.8108	faster
solve-f4	23	232	P	360.9062	32.2386	11.2066	faster
solve-f5	23	232	P	366.3932	32.9762	11.1061	faster
decomp-f4	23	232	P	241.0781	25.3208	9.5504	faster
decomp-f5	23	232	P	243.2470	25.4276	9.5887	faster

Latest native complete-call ms (sample count)

Boundary	Method	n17 planted	n17 generator	n23 planted	n23 generator
complete_roots	roots-gray	0.1757 (5)	0.1752 (5)	2.3277 (5)	2.6233 (5)
complete_roots	roots-wide	0.3535 (5)	0.3420 (5)	1.4074 (5)	1.5652 (5)
complete_roots	solve-f4	0.5217 (5)	0.5308 (5)	32.1783 (5)	34.0685 (5)
complete_roots	solve-f5	0.5240 (5)	0.5350 (5)	32.7056 (5)	33.8614 (5)
complete_roots	solve-inherited	2.8529 (5)	2.8616 (5)	38.6296 (5)	40.8531 (5)
complete_roots	generalized-native	0.5866 (5)	0.6117 (5)	2.6846 (5)	2.7184 (5)
complete_roots	generalized-typed	0.1881 (5)	0.1874 (5)	1.3946 (5)	1.4821 (5)
complete_roots	roots-moebius	0.4946 (5)	0.5007 (5)	29.5689 (5)	35.9190 (5)
complete_roots	roots-monica	0.5060 (5)	0.5054 (5)	12.2638 (5)	12.4504 (5)
complete_roots	generic-gb	30.0242 (5)	29.1847 (5)	645.4938 (5)	461.8308 (5)
decomposition	decomp-f4	1.1100 (5)	1.1397 (5)	7.7178 (5)	36.1335 (5)
decomposition	decomp-f5	1.1176 (5)	1.0993 (5)	7.6194 (5)	36.7864 (5)
decomposition	decomp-f6	1.4737 (5)	1.9511 (5)	7.1367 (5)	18.9605 (5)
decomposition	decomp-f6-pairs	1.4674 (5)	1.9250 (5)	7.0985 (5)	18.6299 (5)
decomposition	decomp-generalized	2.4957 (5)	2.5524 (5)	8.9416 (5)	9.6225 (5)
decomposition	decomp-generalized-typed	0.2126 (5)	0.2116 (5)	1.4523 (5)	1.5887 (5)
decomposition	rootfind-mitm	0.3440 (5)	0.3413 (5)	43.6873 (5)	43.8226 (5)
decomposition	rootfind-table	1.1543 (5)	1.1549 (5)	126.7873 (5)	137.6003 (5)

Four-worker matrix controls

Arm	n	Before ms	After ms	Ratio	Decision
matrix-f4	17	8.3096	8.1812	1.0157	neutral
matrix-f5	17	8.1900	8.1191	1.0087	neutral
matrix-f4	23	60.4794	60.8922	0.9826	neutral
matrix-f5	23	58.3452	58.8872	0.9535	neutral

Instrumented matrix phase medians, ms; separate nonadditive observations

Arm	n	Target	Policy	Build ms	Reduce ms	Readback ms	Calls
solve-f4	17	P	baseline	8.7160	8.9600	0.3806	126.0
solve-f4	17	P	prepared	2.8002	4.6244	0.0315	63.0
solve-f4	17	P	quadratic-tail	0.2880	0.1284	0.0003	1.0
solve-f5	17	P	baseline	8.9412	8.8840	0.3755	126.0
solve-f5	17	P	prepared	2.9434	4.5635	0.0276	63.0
solve-f5	17	P	quadratic-tail	0.2930	0.1251	0.0004	1.0
decomp-f4	17	P	baseline	1.0468	0.9347	0.0384	12.0
decomp-f4	17	P	prepared	0.8150	0.5272	0.0019	6.0
decomp-f4	17	P	quadratic-tail	0.2224	0.1277	0.0003	1.0
decomp-f5	17	P	baseline	1.0859	0.9145	0.0378	12.0
decomp-f5	17	P	prepared	0.8519	0.5370	0.0017	6.0
decomp-f5	17	P	quadratic-tail	0.2261	0.1278	0.0003	1.0
solve-f4	17	G	baseline	9.1018	9.1690	0.3729	126.0
solve-f4	17	G	prepared	2.8679	4.3224	0.0266	63.0
solve-f4	17	G	quadratic-tail	0.2836	0.1171	0.0003	1.0
solve-f5	17	G	baseline	9.3936	9.1950	0.3771	126.0
solve-f5	17	G	prepared	3.0621	4.3303	0.0238	63.0
solve-f5	17	G	quadratic-tail	0.2941	0.1170	0.0003	1.0
decomp-f4	17	G	baseline	3.2501	3.2268	0.1326	44.0
decomp-f4	17	G	prepared	1.3733	1.5879	0.0086	22.0
decomp-f4	17	G	quadratic-tail	0.2250	0.1158	0.0003	1.0
decomp-f5	17	G	baseline	3.3447	3.2641	0.1336	44.0
decomp-f5	17	G	prepared	1.4657	1.5731	0.0081	22.0
decomp-f5	17	G	quadratic-tail	0.2254	0.1143	0.0003	1.0
solve-f4	23	P	baseline	154.4598	185.4734	5.8651	1022.0
solve-f4	23	P	prepared	42.7060	102.8753	0.2799	511.0
solve-f4	23	P	quadratic-tail	11.3662	16.3405	0.0063	63.0
solve-f5	23	P	baseline	158.1360	186.0851	5.8723	1022.0
solve-f5	23	P	prepared	44.1468	102.8984	0.2669	511.0
solve-f5	23	P	quadratic-tail	11.6955	16.4647	0.0045	63.0
decomp-f4	23	P	baseline	10.8251	11.7863	0.3664	64.0
decomp-f4	23	P	prepared	5.1067	7.1176	0.0189	32.0
decomp-f4	23	P	quadratic-tail	2.9963	2.2553	0.0013	7.0
decomp-f5	23	P	baseline	11.0404	11.8074	0.3694	64.0
decomp-f5	23	P	prepared	5.1557	7.1380	0.0171	32.0
decomp-f5	23	P	quadratic-tail	2.9898	2.2184	0.0006	7.0
solve-f4	23	G	baseline	161.0460	188.6339	5.8878	1022.0
solve-f4	23	G	prepared	47.6298	102.8295	0.2699	511.0
solve-f4	23	G	quadratic-tail	12.0189	15.8709	0.0063	63.0
solve-f5	23	G	baseline	167.5921	191.5247	6.0272	1022.0
solve-f5	23	G	prepared	46.9310	98.5088	0.2583	511.0
solve-f5	23	G	quadratic-tail	12.2728	15.8581	0.0042	63.0
decomp-f4	23	G	baseline	164.3466	191.6413	5.9552	1022.0
decomp-f4	23	G	prepared	44.7066	97.7419	0.2533	511.0
decomp-f4	23	G	quadratic-tail	12.0044	16.0784	0.0038	63.0
decomp-f5	23	G	baseline	177.9320	202.0490	6.3354	1022.0
decomp-f5	23	G	prepared	48.9937	103.5476	0.2677	511.0
decomp-f5	23	G	quadratic-tail	12.6099	16.8502	0.0039	63.0